

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 20265
Software Quality Assurance & Testing (CS-468)
Assignment # 1

Semester: VIII

Announced Date: March 12, 2026

Total Marks: 3

Instructor Name: Farhana Ashraf

Batch: 2022F

Due Date: 1st week right after eid Monday to Thursday

Marks Obtained:

Instructions:

- Attempt all parts of the question.
- All parts carry equal parts.
- Submit your **handwritten** assignment in form of hardcopy **in person**. Use **A4** size paper to solve the assignment.

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
CLO 1	Recognize software quality management processes and standards.	PLO_2 (Knowledge for Solving Computing Problems)	C2 (Understanding)

Q1. Answer following questions.

[3 mark]

- (a) It has been claimed that environmental characteristics create the need for intensive and continuous managerial efforts parallel to the professional efforts that have to be invested in order to ensure the project's quality or, in other words, to assure the project's success. **Discuss** the reasons behind this claim, including an analysis of the managerial effort created by each of the SQA environmental characteristics.
- (b) A programmer claims that because only a small proportion of software errors turn into software failures, it is unnecessary to make substantial investments in the prevention and elimination of software errors. Do you agree with this view? If yes, **discuss** the outcome of accepting these views. **Defend** your opinion otherwise.
- (c) George Wise is an exceptional programmer. Testing his software modules reveals very few errors, far fewer than the team's average. He keeps his schedule promptly, and only rarely is he late in completing his task. He always finds original ways to solve programming difficulties, and uses an original, individual version of the coding style. He dislikes preparing the required documentation, and rarely does it according to the team's templates. A day after completing a challenging task, on time, he was called to the office of the department's chief software engineer. Instead of being praised for his accomplishments (as he expected), he was warned by the company's chief software engineer that he would be fired unless he began to fully comply with the team's coding and documentation instructions. Do you agree with the position taken by the department's chief software engineer? Could you **infer** why his or her position was so decisive?